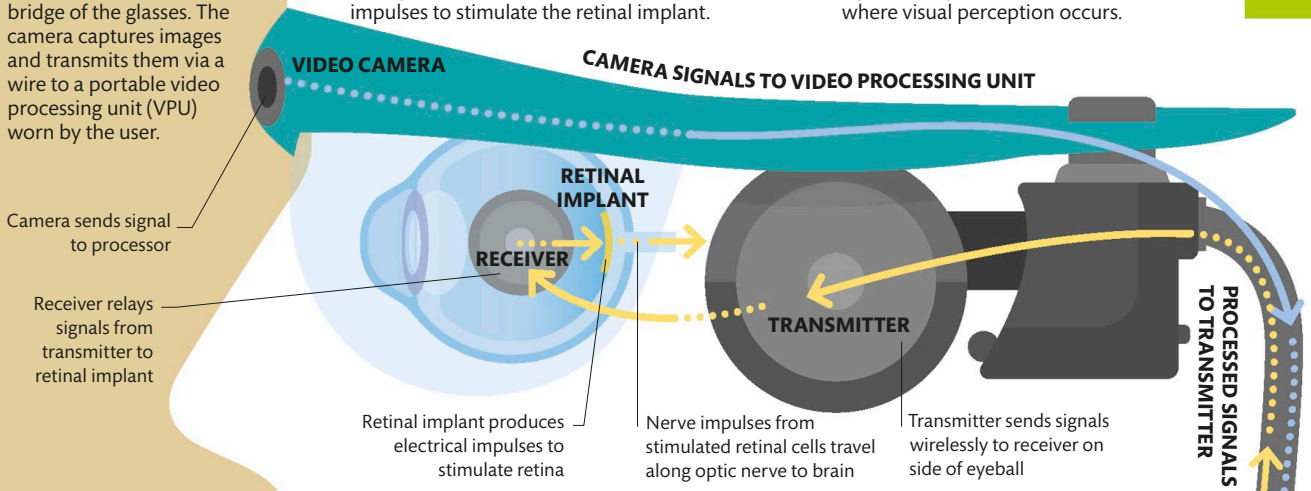
1 Video camera captures scene
The user wears glasses with a miniature video camera fitted to the bridge of the glasses. The camera captures images and transmits them via a wire to a portable video processing unit (VPU) worn by the user.

3 Data transmitted to retinal implant
The transmitter relays signals to a receiver inside the eye socket, on the side of the eyeball. The receiver comprises an antenna, which detects the signals, and an electronic unit, which sends impulses to stimulate the retinal implant.

4 Implant sends data to brain
The implant consists of an electrode array attached to the retina. The electrodes stimulate the retina's remaining cells to send signals along the optic nerve to the brain, where visual perception occurs.

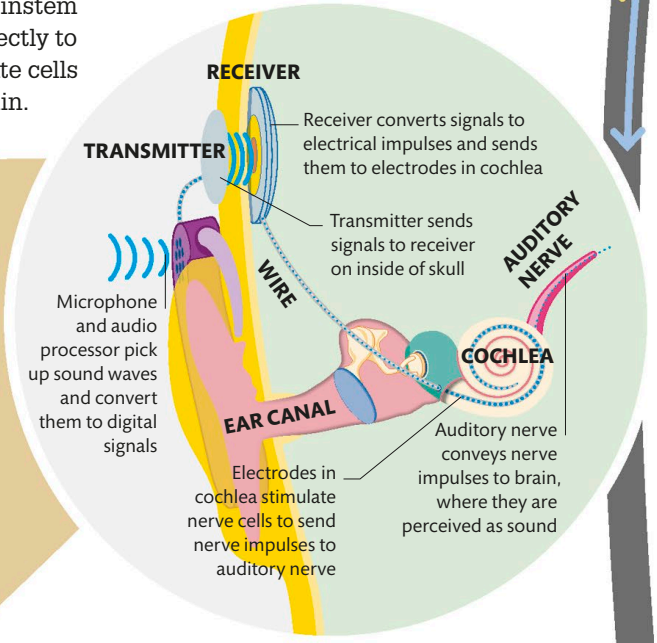


Bionic eye
Damage to cells in the retina (the light-sensitive layer at the back of the eye) can result in loss of vision. Retinal implants, such as the "bionic eye" system, can convert light patterns into data and bypass the damaged retinal cells to send the data to the brain.

Sensory implants

Some brain implants are used to restore vision or hearing in people whose nerves are not sending information efficiently to the brain. Retinal implants can help to restore vision by stimulating the optic nerve to send nerve impulses to the brain. Implants inside the cochlea, in the inner ear, stimulate the auditory nerve to transmit nerve impulses from the inner ear to the brain. If the auditory nerve does not work, an auditory brainstem implant may be fitted directly to the brainstem, to stimulate cells to send signals to the brain.

2 Video data from camera processed
The VPU converts the video signals into a pixellated "brightness map," which it then encodes as digital signals. It sends these signals to a transmitter mounted on the side of the user's glasses.



Cochlear implants
In normal hearing, sound vibrations are transmitted via the eardrums and middle ear bones to the inner ear. Hair cells inside a structure called the cochlea turn these vibrations into electrical signals, which pass along the auditory nerve to the brain. If the internal ear structures are not working properly, an implant can be fitted inside the cochlea to carry signals directly to the auditory nerve.

THE **BATTERIES** USED IN **PULSE GENERATORS** FOR **DEEP BRAIN STIMULATION** LAST UP TO ABOUT **NINE YEARS**